

End of Topic Test – Chapter 3: Modelling and Estimation (Test 2) — Mark Scheme

Total: 47 marks

Note: this test consists mainly of open-ended Fermi-estimation questions. Numerical answers will vary depending on the assumptions made. Award method/accuracy marks for any sensible, clearly stated assumption and consistent, correctly-executed arithmetic, provided the final answer is of a realistic order of magnitude. Example solutions are given below.

1. Fermi estimation: umbrellas on a rainy high street

(a) B1+B1 (any two) reasonable assumptions, e.g. assume the average number of umbrellas per shopper (accounting for some shoppers carrying two, e.g. one per family member, and some carrying none, e.g. those who use a coat or stay indoors); assume the full 40,000 footfall figure represents distinct individuals rather than repeat passes by the same shopper. [2]

(b) M1 assumption quantified, e.g. average ≈ 0.7 umbrellas per shopper. M1 $40,000 \times 0.7$. A1 estimate $\approx 28,000$, i.e. of order 3×10^4 umbrellas (accept any answer in the range roughly 20,000–35,000 with valid supporting method). [3]

(c) B1+B1 (any one, developed) limitation, e.g. umbrella use fluctuates with rainfall intensity throughout the day, so a single average may not reflect this variation; the footfall count may include the same shopper passing the sensor multiple times, which would inflate the estimate. [2]

2. Standard form and scaling

Speed of light 3.0×10^8 m/s; distance to rover 2.25×10^{11} m.

(a) M1 time = distance $\div$ speed = $\frac{2.25 \times 10^{11}}{3.0 \times 10^8}$. A1 = 750 s. [2]

(b) M1 recognise time $\propto$ distance (direct proportion, since $t = d/v$ with v fixed), so tripling distance triples time. M1 new distance = 6.75×10^{11} m. A1 new time = 2250 s. [3]

3. Subdividing: estimating the number of trees in a park

(a) B1+B1 (any two) assumptions, e.g. tree density is uniform across all plots (the sample plot is representative of the whole park); the park boundary is well approximated by the 800 m by 500 m rectangle shown (no significant irregular edges are excluded or extra areas included). B1 a third valid assumption may also be credited if two are not fully developed. [3]

(b) M1 area of one plot: $(800 \div 4) \times (500 \div 5) = 200 \times 100 = 20,000$ m². M1 total number of plots = $4 \times 5 = 20$. M1 total = 20×12 (sample count per plot). A1 estimate = 240 trees (accept alternative sensible readings of the grid with consistent working). [4]

4. Modelling cycle: estimating paper usage in a school

(a) B1+B1+B1 (any three) reasonable, quantified assumptions, e.g. each student uses ≈ 4 sheets per day for printed handouts; each student uses the equivalent of ≈ 2 sheets per day of exercise-book paper; each student wastes ≈ 1 sheet per day through printing/photocopying errors. [3]

(b) M1 printed handouts: $900 \times 4 = 3,600$ sheets. M1 exercise book + wasted sheets: $(900 \times 2) + (900 \times 1) = 1,800 + 900 = 2,700$ sheets. M1 total = $3,600 + 2,700$. A1 total $\approx 6,300$ sheets ($\approx 6.3 \times 10^3$), accept any answer of a similar order of magnitude with valid supporting assumptions. [4]

(c) B2 (or B1+B1) valid improvement, e.g. use actual printer/photocopier meter counts from the school office for a typical week instead of estimated assumptions; account for variation between days (e.g. exam periods or coursework deadlines require more printed sheets). [2]

5. Estimating the volume of a grain silo

Diameter ≈ 2.4 m (radius 1.2 m), height ≈ 6 m.

(a) M1 formula $V = \pi r^2 h$. M1 substitute $r = 1.2$, $h = 6$: $V = \pi \times 1.2^2 \times 6$. A1 $V \approx 27.1 \text{ m}^3$. [3]

(b) B2 (or B1+B1) valid limitation, e.g. a real silo typically has a conical or domed roof and a hopper-shaped base rather than flat ends, so the cylindrical model does not capture the true shape; the wall thickness of the silo reduces the internal usable volume compared with the outer dimensions modelled. [2]

6. Standard form: density of Jupiter

Mass 1.9×10^{27} kg, volume $1.43 \times 10^{24} \text{ m}^3$.

(a) M1 density = mass $\div$ volume. M1 $1.9 \div 1.43 \approx 1.33$ and $10^{27} \div 10^{24} = 10^3$ combined correctly. A1 density $\approx 1.33 \times 10^3 \text{ kg/m}^3$ (accept 1.3×10^3). [3]

(b) B2 (or B1+B1) valid reason, e.g. standard form allows very large or very small numbers to be written, read and compared easily without long strings of zeros; it makes calculations (multiplying/dividing powers of ten) quicker and less prone to error than working with the numbers in full. [2]

7. Assumptions: estimating the number of birds in a nature reserve

Reserve surface area 3.2 km^2 .

(a) B1+B1+B1 (any three) assumptions, e.g. an assumed average bird density per km^2 (a stated numerical value); birds are assumed to be evenly distributed across the reserve; no significant seasonal migration is affecting bird numbers at the time of the estimate. [3]

(b) M1 assumed density stated, e.g. ≈ 650 birds per km^2 . M1 correct multiplication: 3.2×650 . A1 estimate = 2,080 birds (accept any answer of a similar order of magnitude, e.g. 1,500–3,000, with a clearly stated and multiplied assumption). M1 working clearly and logically laid out (assumption $\rightarrow$ calculation $\rightarrow$ estimate). [4]

(c) B2 (or B1+B1) valid improvement, e.g. carry out an actual point-count or transect survey across sample areas of the reserve to estimate density more accurately, then scale up to the whole reserve; use a mark-recapture (ringing) method to estimate the total bird population directly rather than assuming a density. [2]